First Meeting Agenda

Electrical Team Kickoff

NLC STEM Club - Electrical Systems Team

February 2026

1 FIRST ELECTRICAL TEAM MEETING

- Agenda & Leadership Guide
- Fixed-Wing Environmental Monitoring UAV
- NLC STEM Club - Electrical Systems Team
- Meeting Duration: 2.5-3 hours

2 THIS IS A PORTFOLIO-LEVEL PROJECT

- You are not building a hobby drone. You are executing
- aerospace systems engineering work that will demonstrate
- your competency to future employers and graduate programs.
- Act like a systems engineer. Use professional terminology.
- Set the tone for excellence from day one.
- PRE-MEETING PREPARATION (Your Checklist)
- Print Documents (1 copy per attendee + 2 extras):
- This Meeting Agenda (19 pages)
- Power Budget Analysis (10 pages)
- Electrical System Block Diagram (7 pages)
- Detailed Wiring Diagram (16 pages)
- Safety Protocol (11 pages) - SIGNATURES REQUIRED
- Subsystem Assignment Sheet (see Section 9)
- Gather Physical Materials:
- Whiteboard markers (3+ colors for block diagram)
- Component inventory (all electrical parts from parts list)
- Multimeter (for demonstrations)
- OVONIC 4S LiPo battery (fully charged, in safety bag)
- LiPo charger (B6 charger)
- Fire extinguisher (verify location, show to team)
- First aid kit (verify location)
- Pens for signatures

- Prepare Workspace:
- Reserve room for 2.5-3 hours
- Set up tables for component inventory
- Label 4 stations: Power / Flight Electronics / Research / Integration
- Ensure electrical outlets available (for battery charging demo)
- Mental Preparation:
- Review all printed documents thoroughly
- Rehearse opening statement (see Section 2)
- Prepare to answer questions confidently
- Remember: You are the expert. They are looking to YOU for leadership.

3 MEETING STRUCTURE OVERVIEW

4 TIME PHASE OBJECTIVE

0:00 Opening & Context Set professional tone, explain mission
0:15 System Architecture Lecture Teach power flow, design principles
0:45 Critical Technical Decision Decide on Dual-BEC architecture
1:00 Subsystem Organization Assign teams and ownership
1:15 **BREAK** (10 minutes)
1:25 Safety Training Mandatory safety protocol review
1:45 Hands-On Phase 1: Inventory Check components, create tracking sheet
2:15 Hands-On Phase 2: Power Demo Build power test bench, measure voltages
2:35 Week 1-2 Planning Define milestones and deliverables
2:50 Closing & Commitment Confirm next meeting, sign-offs

- SECTION 1: ROOM SETUP (Arrive 30 minutes early)

5 WHITEBOARD LAYOUT:

Title (Top): "ELECTRICAL ARCHITECTURE - Fixed-Wing UAV"

Left Half: Power Flow Block Diagram

(Draw during presentation - see block diagram document for reference)

Battery (14.8V)

↓

ESC + External BEC

↓

Rail 1 (Flight-Critical) — Rail 2 (Servo Power)

Right Half: Key Numbers

- Battery: 14.8V, 2200mAh

- ESC BEC: 5V @ 3A

- External BEC: 5V @ 6A

- Typical Draw: 1383mA

- Peak Draw: 1863mA

Bottom: Design Principles

1. ISOLATION: Flight-critical separate from high-current loads

2. REDUNDANCY: Servo BEC failure doesn't kill FC

3. COMMON GROUND: All systems share ground reference

4. NOISE ISOLATION: Motor wires away from sensors

6 TABLE LAYOUT:

Table 1: Component Inventory & Inspection

- All electrical components laid out
- Labels for each component
- Checklist for verification

Table 2: Documentation Station

- Printed documents (Power Budget, Wiring Diagram, etc.)
- Laptop with Google Sheet for tracking

Table 3: Power Test Bench (Later in meeting)

- Battery (in LiPo bag)
- ESC
- Multimeter
- Flight Controller
- Servos

Table 4: Safety Equipment

- LiPo safety bag
- Fire extinguisher (visible)
- First aid kit
- Safety protocol sign-off sheets
- SECTION 2: OPENING (0:00 - 0:15) - Set the Tone

Stand at front of room. Make eye contact. Speak with confidence.
YOUR OPENING STATEMENT (Memorize or adapt):
"Welcome to the first electrical systems meeting for our UAV project.
Before we start, I want to be clear about what we're doing here. This is not a hobby project. This is not something we're throwing together to fly once and forget about.
We are building a fixed-wing environmental monitoring UAV as a portfolio-level aerospace engineering project. That means we're going to approach this like professional engineers, not hobbyists.

6.1 Today, we will

- Define our electrical architecture
- Make critical engineering decisions
- Assign subsystem ownership
- Begin hands-on validation work

6.2 By the end of this meeting, everyone will understand

- How power flows through this aircraft
- What components you're responsible for
- What risks we need to mitigate
- What we're building in the next 2 weeks

We originally planned a solar-powered system, but due to budget and timeline constraints, we've pivoted to a fixed-wing electric aircraft with a research payload. This is the correct engineering decision, and it actually makes our electrical systems more robust.
Let me be direct: electrical is the critical path for this project. If we get this wrong, nothing else matters. Mechanical can build the perfect airframe, software can write perfect code, but if we have a power failure at altitude, we crash.
So we're going to do this right. We're going to use professional terminology, follow systematic testing procedures, and prioritize safety above all else.
Questions before we dive into the architecture?"

7 LEADERSHIP TONE TO ESTABLISH:

- Use these phrases:
- "Let me walk you through the architecture..."
- "From a systems engineering perspective..."
- "The failure mode here is..."
- "We need to validate this through testing..."
- "This gives us engineering margin because..."
- "Isolation prevents this failure from cascading..."
- Avoid these phrases:
- "I think maybe we could..."
- "I'm not sure, but..."
- "Whatever you guys want to do..."
- "This is just a club project..."
- "We'll figure it out later..."

You are the expert. They need you to be confident and clear.

- SECTION 3: SYSTEM ARCHITECTURE LECTURE (0:15 - 0:45) - Teach

OBJECTIVE: Every team member understands power flow and design principles.
PART 3A: Draw the Block Diagram (15 minutes)
Start at the top of whiteboard. Draw as you explain. Use different colors.
"Let's start at the source of all power: our OVONIC 4S LiPo battery."
[Draw battery at top]
"4S means 4 cells in series. Each cell is 3.7V nominal, so:

- 3.7V $\times$ 4 = 14.8V nominal voltage
- Fully charged: 4.2V per cell = 16.8V total

- Empty: 3.3V per cell = 13.2V total (never go below this - battery damage)
- Capacity: 2200mAh = 2.2Ah
- This battery can deliver over 286 amps burst (we won't, but it can)"

[Draw line down from battery, split into two paths]
"Power splits into two main paths:
Path 1: The ESC - Electronic Speed Controller"
[Draw ESC on left]
"This does two things:

- 1. Provides high-power 14.8V to the motor (10-30 amps)
- 2. Has a built-in BEC - Battery Eliminator Circuit - that gives us 5V @ 3A"

[Draw motor below ESC]
"Motor draws the most power - 10 to 18 amps typical, up to 30 amps full throttle.
That's 14.8V $\times$ 18A = 266 watts of power. This is why the motor dominates our power budget."
[Draw BEC output coming from ESC]
"The ESC's BEC gives us a 5V rail for electronics. 5V at 3A maximum.
This is Rail 1 - our Flight-Critical rail."
[Draw Rail 1 components: Flight Controller, GPS, Receiver, Arduino]
"Rail 1 powers:

- MATEK F405-WMN Flight Controller: The brain. 250mA typical.
- BN-220 GPS: Navigation. 60mA.
- ELRS Receiver: Radio link to ground station. 90mA.
- Arduino Nano: Research payload brain. 30mA.
- Sensors: BME280 and MQ135. 160mA combined."

[Write currents next to each component, sum at bottom]
"Total: 663mA. We have 3000mA available from the BEC.
That leaves 2337mA margin. That's 78% headroom. Excellent, right?"
[Pause for effect]
"Wrong. Because I haven't told you about the servos yet."
[Draw 4 servos]
"We have 4 servos:

- 2 ailerons (roll control)
- 1 elevator (pitch control)
- 1 rudder (yaw control)

A single servo draws:

- 10mA idle (just sitting there)

- 180mA moving (normal operation)
- 300mA peak (fast movement)
- 650mA stalled (control surface blocked or jammed)

In flight, all 4 servos move constantly. That's 4 × 180mA = 720mA typical, 1200mA peak."
[Write on whiteboard]
"663mA (electronics) + 720mA (servos) = 1383mA total typical draw.
We have 3000mA. Still okay, right? 1617mA margin."
[Write in red]
"But what if a servo stalls? Imagine a control surface gets jammed.
One servo tries to move, can't, draws 650mA.
Now: 663 + 650 + (3 × 180) = 1853mA. Still under 3A.
But what if TWO servos stall? Or THREE?
Now we're at 2500mA or more. We've exceeded the BEC capacity."
[Draw X through the circuit]
"What happens when you try to draw more current than a BEC can provide?

- 1. Voltage sags - drops below 5V
- 2. Flight controller browns out - resets
- 3. You lose control
- 4. You crash"

[Let that sink in for 3 seconds]
"This is called a single point of failure. Everything depends on one BEC."
PART 3B: Introduce the Solution (10 minutes)
[Draw second path from battery]
"The solution: Add a second BEC. We call this Dual-BEC Architecture."
[Draw External UBEC on right side of diagram]
"External UBEC:

- Input: 14.8V from battery (via XT60 splitter)
- Output: 5V @ 6A (twice the capacity of ESC BEC)
- This becomes Rail 2: Servo Power Rail"

[Draw servos connected to Rail 2 for power]
[Draw servo signal wires going to Flight Controller]
"Now the servos get power from Rail 2, signal from Flight Controller.
This is called power/signal separation. It's standard aerospace practice."
[Write on whiteboard]
RAIL 1 (ESC BEC, 3A): RAIL 2 (External BEC, 6A):

- Flight Controller - 4× Servos
- GPS

- Receiver
- Arduino + Sensors
- Total: 663mA Total: 720-1200mA
- Margin: 2337mA (78%) Margin: 4800mA (80%)

"Now, even if ALL FOUR servos stall simultaneously ($1 \times 650 + 3 \times 300 = 1550$mA), we still have margin on Rail 2, and Rail 1 is completely unaffected.
Flight Controller voltage: STABLE.
GPS lock: MAINTAINED.
Control link: ACTIVE.
If Rail 2 fails completely, the flight controller stays powered. We can't move control surfaces, but we can glide to a controlled landing.
If Rail 1 fails, servos keep working, but we lose the brain - catastrophic.
So we protect Rail 1 at all costs."
[Circle on whiteboard]
"This is ISOLATION. High-current loads cannot affect flight-critical systems."
PART 3C: Design Principles (5 minutes)
[Write on whiteboard, explain each]
"Five design principles guide our electrical architecture:

8 1. ISOLATION

- Flight-critical systems are isolated from high-current loads.
- Servo power rail is separate from FC power rail.

9 2. REDUNDANCY

- If servo BEC fails, FC stays alive.
- If FC fails... we're in trouble, so we protect FC at all costs.

10 3. COMMON GROUND

- All systems share the same ground reference.
- Battery negative is the master ground for everything.
- This prevents ground loops and signal noise.

11 4. NOISE ISOLATION

- Motor wires carry high-frequency noise from ESC switching.

11.1 Keep motor wires AWAY from

- GPS antenna
- Receiver antenna
- Analog sensors (MQ135)
- Twist motor wires together to reduce electromagnetic interference.

12 5. POWER MARGIN

- We size everything with headroom.
- If a BEC can provide 3A, we only use 1.4A typical.
- This accounts for component tolerance, aging, and unexpected spikes.
- Real aerospace systems use 50-100% margin. We're doing the same."

[Turn to team]
"Questions on the architecture before we make our technical decision?"
[Take questions. This is important - make sure everyone understands before moving forward]

- SECTION 4: CRITICAL TECHNICAL DECISION (0:45 - 1:00) - Decide

OBJECTIVE: Team commits to Dual-BEC architecture (Option B).
"We have a decision to make. I'm going to present three options, then we'll discuss and vote."
[Refer to Power Budget document, Section 5]
OPTION A: Single BEC Configuration
"Use ESC BEC for everything. Simplest wiring.

12.1 Pros

- Simple
- Minimal wiring
- One less component

12.2 Cons

- Only 1137mA margin under peak load (38%)
- Servo stall can exceed BEC capacity
- Single point of failure
- Risk of brownout → FC reset → crash
- No engineering margin

My assessment: UNSAFE for portfolio-level aerospace work.
This is acceptable for a toy quadcopter. Not for a serious UAV."
OPTION B: Dual-BEC Configuration (RECOMMENDED)
"Add external 5V 6A UBEC for servos. Professional power distribution.

12.3 Pros

- Rail 1: 78% margin (flight-critical systems protected)
- Rail 2: 80% margin (servos can handle simultaneous stall)
- Isolation: Servo load cannot affect FC
- Redundancy: Servo BEC failure doesn't kill FC
- Professional aerospace practice
- Strong portfolio demonstration

12.4 Cons

- Slightly more complex wiring
- Need to purchase external BEC (~$10-15)
- Additional component to integrate

Cost: $15 for UBEC
Benefit: Insurance against mid-flight power failure and loss of $500 aircraft
My assessment: This is the CORRECT engineering choice.
This is what I would do if this were my job at Lockheed Martin or Boeing."
OPTION C: Hybrid Configuration
"Similar to Option B, but Arduino powered from Rail 2 instead of Rail 1.
This is acceptable if you want maximum isolation. Research payload failure cannot affect flight systems. But adds complexity for marginal benefit."

13 FACILITATE DISCUSSION

"Thoughts? Concerns? Devil's advocate arguments?"
[Let team discuss for 5 minutes]

13.1 Common questions you might get

Q: "Do we really need the extra margin?"
A: "In engineering, you don't size for typical case. You size for worst case. What if a control surface freezes due to icing? What if a servo fails and draws excessive current? Margin protects us when unexpected things happen. And unexpected things ALWAYS happen."
Q: "Can we just test with one BEC first and add second later if needed?"
A: "No. If we test with insufficient power architecture and encounter brownouts during testing, we've just made our debugging 10× harder. Are bugs due to software? Wiring? Configuration? Or just insufficient power? We eliminate power as a variable by doing it right from the start."
Q: "How much will the external BEC cost?"
A: "$10-15 on Amazon. Hobbywing 5V 6A UBEC is $12. Given our total project budget is $500, this is 2.4% additional cost for significant risk mitigation. Worth it."

14 VOTE & COMMIT

"Let's vote. Show of hands:
Option A (Single BEC): [Should be 0-1 hands]
Option B (Dual BEC): [Should be majority]
Option C (Hybrid): [Maybe 0-2 hands]"

14.1 If vote is not unanimous for Option B, use this

"I hear the concerns about added complexity. But fundamentally, this is a question of risk tolerance. Option A has a ¿50% chance of causing brownout during aggressive maneuvering based on the power budget calculations. That's unacceptable.
As Electrical Lead, my recommendation is Option B. This is my formal engineering decision. Does anyone have a SAFETY or TECHNICAL concern that would override this?"
[Address any technical concerns. If none:]
"Then we're implementing Option B: Dual-BEC architecture. I'll order the external UBEC today. It'll arrive in 2-3 days, and we'll integrate it in Week 1."
[Write on whiteboard in large letters:]

15 DECISION: DUAL-BEC ARCHITECTURE (OPTION B) ✓

"This is now our baseline electrical architecture. All wiring diagrams, testing procedures, and documentation will reflect this design."

- SECTION 5: SUBSYSTEM ORGANIZATION (1:00 - 1:15) - Organize

OBJECTIVE: Assign every team member to a subsystem with clear ownership.
"Electrical has 4 subsystems. Each needs an OWNER and 1-2 assistants.

15.1 The owner's job

- Become the expert on that subsystem
- Create the wiring harness or assembly
- Test and validate functionality
- Document issues and resolutions
- Present status at each meeting

This is your section of the electrical report. You own it."

16 SUBSYSTEM 1: POWER TEAM

16.1 Scope

- Battery management (charging, storage, safety)
- XT60 connections and wiring

- ESC installation and wiring
- External BEC installation
- Power distribution board/harness (Rail 1 and Rail 2)
- Current sensing setup
- Power switch implementation

16.2 Skills Needed

- Soldering (XT60 connectors, power distribution)
- Multimeter usage (measure voltages, current, continuity)
- Attention to safety (LiPo handling)

16.3 Deliverables (Week 1-2)

- LiPo battery charged and tested (voltage check, balance check)
- XT60 Y-harness or splitter built (Battery → ESC + External BEC)
- External UBEC wired and tested (5V output verified)
- Power distribution harness (Rail 1 and Rail 2 wiring)
- Current sensing connected to FC
- All power connections labeled and heat-shrunk
- Power budget validation report (measure actual currents vs predictions)

Owner: _____ Assistant(s): _____

17 SUBSYSTEM 2: FLIGHT ELECTRONICS TEAM

17.1 Scope

- MATEK F405-WMN flight controller setup
- ESC signal wiring to FC
- Servo signal wiring to FC (PWM outputs)
- GPS module (BN-220) wiring and configuration
- ELRS receiver (EP2) binding and wiring
- Flight controller firmware configuration (iNav or ArduPilot)
- Servo calibration (endpoints, direction, center points)

17.2 Skills Needed

- Flight controller software (iNav Configurator or Mission Planner)
- Firmware flashing (if needed)
- Servo calibration procedures
- Radio binding procedures (ELRS)

17.3 Deliverables (Week 1-2)

- Flight controller powered and communicating via USB
- Firmware flashed (iNav or ArduPilot - team decision)
- GPS module wired, confirmed GPS lock (satellite count ¿8)
- ELRS receiver bound to transmitter (Jumper Smart ELRS TX)
- Receiver channels configured (verify stick inputs in FC software)
- ESC signal wire connected, motor direction verified (NO PROP, bench test)
- All 4 servos connected and responding to transmitter inputs
- Servo directions verified/reversed as needed (control surface check)
- Failsafe configured (motor cuts, servos hold last position or center)

Owner: _____ Assistant(s): _____

18 SUBSYSTEM 3: RESEARCH PAYLOAD TEAM

18.1 Scope

- Arduino Nano setup and programming
- BME280 sensor (temp/humidity/pressure) wiring and testing
- MQ135 sensor (air quality) wiring and testing
- MicroSD module wiring and data logging implementation
- Toggle switch integration (logging control)
- Data logging code development (CSV format)
- MQ135 warm-up procedure (24-48 hour burn-in before flight)

18.2 Skills Needed

- Arduino programming (C++)
- I2C communication (BME280)
- Analog input reading (MQ135)
- SPI communication (MicroSD)
- File I/O (SD card library)
- Sensor calibration

18.3 Deliverables (Week 1-2)

- Arduino Nano powered from 5V rail (Rail 1 or Rail 2)
- BME280 connected via I2C (A4/A5), test sketch confirms readings
- MQ135 connected to A0, analog readings logged to Serial Monitor
- MicroSD module connected via SPI (D10-D13), card detected and writable
- Toggle switch wired to D2 with internal pullup resistor
- Integrated data logging code:
- Reads BME280 (temp, humidity, pressure)
- Reads MQ135 (analog gas level)
- Timestamps each reading
- Writes CSV file to SD card when switch is ON
- File format: timestamp,temp,humidity,pressure,gas_level
- MQ135 burn-in completed (powered continuously for 48 hours before first flight)
- Bench test: 5-minute logging session, CSV file retrieved and verified

Owner: _____ Assistant(s): _____

19 SUBSYSTEM 4: INTEGRATION & TESTING TEAM

19.1 Scope

- Oversee integration of all 3 subsystems
- Continuity testing (verify all ground connections)
- Voltage measurements at all power rails
- Current draw measurements (clamp meter or inline sensor)

- Electrical noise testing (GPS lock during motor operation)
- Wire management (routing, zip ties, strain relief)
- Vibration isolation for sensitive components (GPS, FC, sensors)
- Final pre-flight electrical checks

19.2 Skills Needed

- Multimeter expertise (voltage, current, continuity, resistance)
- Systems thinking (how subsystems interact)
- Troubleshooting methodology
- Attention to detail

19.3 Deliverables (Week 1-2)

- Continuity check: All grounds connected (0Ω resistance from battery negative to
- all component grounds: ESC, FC, Arduino, servos, sensors)
- Voltage measurements logged:
- Battery voltage: 14.0-16.8V
- Rail 1 voltage: 5.0V ± 0.1V (no load and under load)
- Rail 2 voltage: 5.0V ± 0.1V (no load and under load)
- Current measurements logged:
- Rail 1 idle: ˜320-400mA ✓
- Rail 1 with servos: ˜1000-1400mA ✓
- Rail 2 idle: ˜40mA ✓
- Rail 2 with servos moving: ˜700-1200mA ✓
- Motor current: 10-18A typical (with clamp meter)
- Voltage sag test: Measure FC voltage during aggressive servo movement
- (must stay ¿4.8V, ideally ¿4.9V)
- Electrical noise test: GPS lock maintained during motor spin test (50% throttle,
- NO PROP, satellite count must not drop)
- Wire routing inspection: No wires rubbing against control surfaces, sharp edges,
- or moving parts
- Vibration isolation: GPS and FC mounted with foam or rubber dampers
- Final pre-flight checklist created (see Wiring Diagram Section 6)

Owner: _____ Assistant(s): _____

20 ASSIGNMENT PROCESS

"Let me explain what each subsystem does, then we'll assign teams.
[Explain each subsystem - read the Scope section above]

20.1 Now, think about

- What skills do you have? (Soldering, programming, testing)
- What do you want to learn?
- Who do you work well with?

I need 1 owner per subsystem. Owners must commit to:

- Attending all meetings
- Spending 4-6 hours per week outside meetings on your subsystem
- Being the go-to expert for your area

Who's interested in each subsystem?"
[Facilitate assignment. Aim for 2-3 people per subsystem depending on team size]
[Fill out Subsystem Assignment Sheet - see Section 9]
"Great. Owners, you are now accountable for your subsystem. Next meeting, I expect status updates from each of you. If you hit a blocker, you escalate to me immediately. Don't wait until the next meeting to tell me you're stuck."

- *BREAK** (1:15 - 1:25) - 10 Minutes

"Let's take a 10-minute break. When we come back, we'll do mandatory safety training, then hands-on work. Stretch, grab water, use the restroom."

- SECTION 6: SAFETY TRAINING (1:25 - 1:45) - Critical

OBJECTIVE: Every team member understands hazards and commits to safety procedures.
[Stand next to safety equipment - LiPo bag, fire extinguisher, first aid kit]
"Before we touch any hardware, we're doing mandatory safety training.

20.2 This project involves three main hazards

1. LiPo batteries - fire/explosion risk
2. Propellers - laceration risk
3. Electrical - burns from soldering, short circuits

I'm going to cover each one, then you'll sign the safety protocol acknowledging you understand the risks. If you're not willing to sign, you cannot participate in hands-on work. No exceptions."
LIPO BATTERY SAFETY (10 minutes)
[Hold up LiPo battery IN safety bag]
"This is an OVONIC 4S 2200mAh LiPo battery. It stores 32.56 watt-hours of energy.
If mishandled, it can enter thermal runaway - an uncontrollable chemical reaction

that burns at over 1000°C. LiPo fires cannot be extinguished with water. They burn for 10-30 minutes and release toxic smoke.
Google 'LiPo fire' after this meeting. Watch a few videos. Take this seriously.
[Refer to Safety Protocol document, Section 1]

20.3 Key rules

NEVER charge unattended. NEVER.

- If you're charging and need to leave, stop charging, disconnect, store in bag.

NEVER discharge below 3.3V per cell.

- Our battery management will cut power at 3.5V/cell (14.0V total).

NEVER overcharge above 4.2V per cell.

- The B6 charger handles this automatically, but still monitor.

NEVER use a swollen or damaged battery.

- If you see swelling or damaged wires, report immediately. Do not charge or use.

ALWAYS store in LiPo safety bag.

- This bag contains fire if battery ignites. It's not perfect, but it's critical.

ALWAYS charge on non-flammable surface.

- Metal, ceramic, concrete. NOT wood, carpet, or plastic.

[Demonstrate LiPo charger]
I'll now show you the proper charging procedure.
[Walk through B6 charger setup - see Safety Protocol]
1. Inspect battery (no swelling, no damage)
2. Place in safety bag on metal surface
3. Connect balance plug (5-pin JST-XH)
4. Connect main power (XT60)
5. Set charger: LiPo Balance Charge, 4S, 1.0A charge rate
6. Press START, verify 4S 14.8V, press START again
7. Monitor during entire charge (60-120 minutes)
8. When done, disconnect balance plug first, then main plug
9. Let cool 15 minutes before use

20.4 If battery becomes hot (¿50°C), smells unusual, or swells during charging

- STOP IMMEDIATELY. Disconnect. Move outdoors if safe. Monitor for 1 hour.

[Point to fire extinguisher]
Fire extinguisher is here. ABC rated, works on LiPo fires (barely). If LiPo catches fire, evacuate, call 911, let it burn out outdoors if possible.
Questions on LiPo safety?"
[Take questions]
PROPELLER SAFETY (5 minutes)
"A 10-inch propeller at full throttle has tip speed around 250 mph. Contact causes severe lacerations or amputation. This is not an exaggeration.

20.5 Key rules

REMOVE propeller for ALL bench testing indoors.

- We will not install the propeller until Week 2 outdoor testing.

NEVER test motor with propeller indoors.

- No matter how careful you think you'll be. NO PROP INDOORS.

NEVER reach toward spinning propeller.

- Obvious, but people make mistakes. Stay behind the plane of rotation.

20.6 When we do outdoor propeller tests

- 10-meter safety perimeter
- Safety officer monitors perimeter
- Verbal warning before arming motor: 'PROPELLER TEST - CLEAR THE AREA'
- Wait 10 seconds for confirmation
- Operator stands behind aircraft

Questions on propeller safety?"
ELECTRICAL SAFETY (5 minutes)
[Show soldering iron]
"Soldering iron tip reaches 350°C (660°F). Contact causes instant severe burns.

20.7 Rules

- Always use soldering iron stand
- Wear safety glasses (solder can splatter)
- Work in ventilated area (flux fumes)
- Wash hands after soldering (lead exposure)
- Unplug immediately after use

[Show multimeter]
"Multimeter safety:

- Red probe to positive, black to negative/ground
- Use correct setting (voltage vs current)
- Don't probe live circuits above 50V without training

[Hold up XT60 connector]
"Short circuit prevention:

- Our battery can deliver 286 amps in a short circuit
- That's enough to melt wires instantly and start a fire
- ALWAYS disconnect battery before wiring work
- Use heat shrink tubing on all solder joints
- Keep metal tools away from exposed battery terminals

Questions on electrical safety?"

21 SAFETY PROTOCOL SIGN-OFF

[Hand out Safety Protocol document]
"Everyone read Section 7: Safety Acknowledgment. Page 11.
If you understand and agree to follow these safety procedures, sign the acknowledgment.
If you're not willing to commit to these procedures, you can contribute to this project through CAD design, documentation, or analysis, but you cannot participate in hands-on electrical work.
Take 2 minutes to read and sign."
[Collect signed acknowledgments. File these - they're important if an incident occurs]
"Thank you. Safety is everyone's responsibility. If you see unsafe behavior, speak up.
I don't care if it's me doing something unsafe - call it out. No exceptions."

- SECTION 7: HANDS-ON PHASE 1 - INVENTORY (1:45 - 2:15) - Inspect

OBJECTIVE: Verify all components present and functional. Create tracking system.
"Now we get hands-on. First task: verify we have all components and they're functional.
We're going to create an inventory spreadsheet. This is professional aerospace practice.
Every component is logged, inspected, and tracked."
[Open laptop, create Google Sheet, share with team]

22 SPREADSHEET COLUMNS:

— Component — Qty — Voltage — Current — Connector — Condition — Tested? — Assigned To — Notes —
[Lay out all components on Table 1]
"Each subsystem team, go through your components:

22.1 Power Team

- OVONIC 4S LiPo battery

23 - 40A ESC

- B6 Battery Charger
- LiPo Safety Bag
- XT60 connectors (need to verify if Y-harness included or needs to be built)

23.1 Flight Electronics Team

- MATEK F405-WMN Flight Controller
- BN-220 GPS Module
- ELRS EP2 Receiver
- 4× RGBZONE ES08MA II Servos
- FLASH HOBBY D2830EVO Motor
- Folding Propeller (1060 or 1160)

23.2 Research Payload Team

- Arduino Nano
- BME280 Sensor Module
- MQ135 Air Quality Sensor
- MicroSD Card Module
- SanDisk 32GB MicroSD Card
- Toggle Switch (SPST, 2-pin)

23.3 Integration Team

- Multimeter (verify we have one)
- Soldering iron kit
- Wire (various gauges)
- Heat shrink tubing
- XT60 connectors (spares for power distribution)
- Zip ties, foam tape, etc.

23.4 For each component

1. Verify physical condition (no visible damage)
2. Check connector types (matches wiring diagram?)
3. Identify any missing components or connectors
4. Log in spreadsheet

You have 20 minutes. If you find issues, report to me immediately."
[Circulate among teams, answer questions, verify inventory]

24 MISSING COMPONENTS DISCUSSION

[After 20 minutes, reconvene]
"What are we missing?"
[Expected: External 5V UBEC, possibly XT60 Y-harness, possibly some connectors/wire]
"I'll order [missing items] today. They'll arrive in 2-3 days. In the meantime, we can work on subsystems that don't require those parts."
[Update spreadsheet with missing items and expected arrival dates]

- SECTION 8: HANDS-ON PHASE 2 - POWER DEMO (2:15 - 2:35) - Validate

OBJECTIVE: Build power test bench. Verify ESC BEC output. Demonstrate measurements.
"Power Team, you're up. We're going to build a simple test bench to verify the ESC BEC output. Everyone else, watch and learn - you'll be doing similar tests with your subsystems."
[Clear Table 3 for power test bench]
STEP 1: Battery Voltage Check
[Multimeter in DC voltage mode, 20V range]
"First, we verify battery voltage. Red probe to positive, black to negative."
[Measure across XT60 terminals]
"Reading: _____V [Write on whiteboard]
Expected: 14.0-16.8V
Acceptable: ¿14.5V (battery has charge)
If ¡14.0V: Battery needs charging before proceeding
Our battery is at ____V, which is [good/needs charging]."
[If needs charging, demonstrate proper charging procedure per Safety Protocol]
STEP 2: ESC BEC Output Test (NO MOTOR CONNECTED)
"Now we'll verify the ESC's BEC output. Important: NO MOTOR connected yet. We're only testing the BEC function."
[Connect battery to ESC input (XT60)]
"ESC is now powered. Listen for beeps - that's the ESC initializing."
[Take ESC's BEC output cable (red/black servo connector)]
"This is the BEC output. Red is +5V, black is ground."
[Multimeter: red probe to red wire, black probe to black wire]
"Measuring BEC output voltage..."
[Read voltage]
"Reading: _____V [Write on whiteboard]
Expected: 5.0V ± 0.1V
Acceptable range: 4.9-5.1V
If outside this range, ESC may be faulty.
Our BEC output: ____V ✓"
[Disconnect battery]
"This confirms Rail 1 will have stable 5V. Good."
STEP 3: Servo Movement Test (NO SIGNAL)
"Now let's test a servo. We'll connect power only, no signal."
[Connect one servo's red wire to BEC +5V (red), brown wire to BEC GND (black)]
[Do NOT connect servo signal wire yet]
"Servo is now powered. Move the servo arm by hand."
[Manually rotate servo arm]

"Feel that resistance? That's the servo motor. It should move smoothly with some resistance.
If it's completely stiff, servo is locked up - bad.
If it's completely loose, servo gears are stripped - bad.
This servo: [good/bad]"
[Test all 4 servos this way, log in spreadsheet]
STEP 4: Current Measurement Demo (If time permits)
"If we have a clamp meter or inline current sensor, I'll show you how to measure current draw. This is how we'll validate our power budget later."
[If equipment available, demonstrate measuring current from BEC during servo movement]
"When we do full system integration next week, we'll measure:

- Rail 1 current (should be ~320mA idle, ~660mA with Arduino active)
- Rail 2 current (should be ~40mA idle, ~720mA with servos moving)
- Battery current (should be 10-18A with motor running at cruise throttle)

This validates our power budget analysis."
[Disconnect everything, store battery in LiPo bag]
"Good work, Power Team. You've validated the first part of our power system."

- SECTION 9: WEEK 1-2 PLANNING (2:35 - 2:50) - Plan

OBJECTIVE: Define milestones for next 2 weeks. Assign deliverables.
"Let's define what we're accomplishing in the next 2 weeks. We're in Validation Phase.
That means we validate each subsystem individually BEFORE integrating into the airframe."
[Write on whiteboard]
WEEK 1 MILESTONES (Next 7 Days)

24.1 Power Team

- External UBEC arrives and tested (5V output verified)
- XT60 Y-harness or splitter built (Battery → ESC + UBEC)
- Power distribution harness designed (Rail 1 and Rail 2 wiring plan)
- All power connections soldered and heat-shrunk
- Power bench test: ESC + UBEC both outputting stable 5V simultaneously

24.2 Flight Electronics Team

- Flight Controller connected to computer via USB
- Firmware flashed (team decides: iNav or ArduPilot - recommend iNav for fixed-wing)
- GPS module wired to FC, GPS lock confirmed outdoors (¿8 satellites)
- Receiver bound to transmitter, channels configured in FC software
- Receiver channels tested: move TX sticks, verify response in FC config tool

24.3 Research Payload Team

- Arduino Nano powered via USB, blink sketch loaded
- BME280 sensor wired and tested (I2C address found, readings valid)
- MQ135 sensor wired and tested (analog readings logged to Serial Monitor)
- MicroSD module wired, SD card detected, test file written successfully
- Basic data logging code written (reads sensors, prints to Serial)

24.4 Integration Team

- Multimeter training completed (all team members can measure V, A, continuity)
- Continuity of grounds verified across all subsystems when assembled
- Testing checklist created for Week 2 full system integration
- Wire routing plan drafted (keep motor wires away from GPS/sensors)

WEEK 2 MILESTONES (Days 8-14)

24.5 Power Team

- Full power system bench test: Battery → ESC + UBEC → Rail 1 + Rail 2
- Current measurements logged for Rail 1 and Rail 2 under load
- Voltage sag test: measure FC voltage during servo movement (must be ¿4.8V)

24.6 Flight Electronics Team

- ESC signal wire connected to FC, motor direction verified (NO PROP, bench only)
- All 4 servos connected to FC PWM outputs, signal wires tested
- Servo configuration in FC: direction, endpoints, center points
- Full control surface check: move TX sticks, verify correct servo response
- Failsafe configured and tested (turn off TX, verify motor cuts, servos safe)

24.7 Research Payload Team

- Integrated data logging code complete:
- Toggle switch controls logging on/off
- CSV file written to SD card
- Timestamps, temp, humidity, pressure, gas level all logged
- MQ135 burn-in started (powered continuously for 48 hours before first flight)
- 5-minute bench test: log data, retrieve SD card, verify CSV file readable

24.8 Integration Team

- Full system integration: All subsystems powered simultaneously
- System-level current draw measured and logged
- GPS lock verified with motor running (50% throttle, NO PROP)
- All wires routed and secured (zip ties, strain relief)
- Components vibration-isolated (GPS and FC on foam/rubber)
- Pre-flight electrical checklist completed

25 DELIVERABLE FOR WEEK 2 MEETING

25.1 By next meeting (in 1 week)

25.2 Each subsystem owner brings

- 5-minute status presentation
- What you accomplished
- What you tested
- What worked
- What didn't work
- What you learned
- Blockers (if any)
- Photos/videos of your work
- Show your wiring
- Show test results
- This documents the project for your portfolio
- Updated component spreadsheet
- Mark your components as "Tested" when validated

25.3 By Week 2 meeting

- FULL SYSTEM demonstration ready:
- All subsystems powered and functional on bench
- Data logging working
- Servos responding to transmitter
- GPS locked
- Ready for airframe integration

26 MEETING SCHEDULE

Next Meeting: [DATE, TIME, LOCATION]
Week 2 Meeting: [DATE, TIME, LOCATION]

26.1 Between meetings

- Work on your subsystem 4-6 hours per week
- Communicate in team group chat (set up Discord/Slack/GroupMe)
- If you hit a blocker, message me IMMEDIATELY - don't wait until next meeting
- Document everything (photos, notes, measurements)
- SECTION 10: CLOSING (2:50 - 3:00) - Commit

27 YOUR CLOSING STATEMENT

"That's our first electrical meeting. Let me summarize what we accomplished:

1. ✓ Defined electrical architecture: Dual-BEC power distribution
2. ✓ Made critical engineering decision: Option B architecture
3. ✓ Assigned subsystem ownership: Power, Flight, Research, Integration
4. ✓ Completed safety training: Everyone signed safety protocol
5. ✓ Validated component inventory: We know what we have and what we need
6. ✓ Demonstrated power test bench: Verified ESC BEC works
7. ✓ Defined Week 1-2 milestones: Clear deliverables for each team

This was a productive meeting. But the meeting is the easy part. The work happens between meetings.

27.1 You now own your subsystem. You are responsible for

- Learning how your components work
- Testing thoroughly before integration
- Documenting your work
- Delivering by Week 2

This is not a participation trophy project. This is real engineering work with real deliverables and real deadlines. If you commit, I will support you 100%. If you don't have time, tell me now so we can adjust team assignments.

27.2 Looking ahead

- Week 1-2: Subsystem validation (what we just planned)
- Week 3-4: Full system integration and bench testing
- Week 5-6: Airframe integration and ground tests

- Week 7-8: Flight testing and data collection
- Week 9-10: Analysis and final report

We're on track. You have everything you need to succeed.
Questions? Concerns? Confusion about your subsystem responsibilities?"
[Take questions]
"Alright. Let's make this the best aerospace project NLC STEM Club has ever done.
Take your assignment sheets. Take your printed documents. Study the wiring diagram and power budget analysis.
I'll see you at the next meeting with progress updates.
Go build something great."

- POST-MEETING ACTIONS (Your To-Do List)

27.3 Within 24 hours

- Order missing components (External UBEC, connectors, etc.)
- Set up team communication channel (Discord, Slack, GroupMe)
- Send meeting summary to all attendees + faculty advisor
- Upload component inventory spreadsheet (Google Sheets, shared access)
- File safety protocol sign-off sheets (scan and keep copies)

27.4 Within 48 hours

- Follow up with each subsystem owner individually:
- "How are you feeling about your subsystem? Any questions?"
- Review power budget analysis again (you'll be asked questions)
- Watch YouTube tutorials on:
- iNav or ArduPilot setup (whichever team chose)
- ELRS receiver binding
- Arduino sensor interfacing

27.5 Weekly

- Check in with each subsystem owner via group chat
- Monitor progress on milestones
- Address blockers immediately
- Update project documentation

27.6 Before next meeting

- Prepare meeting slides (status updates, agenda for Meeting #2)
- Test any newly arrived components (External UBEC, etc.)
- Review subsystem deliverables to ensure they're on track

28 APPENDICES

Appendix A: Subsystem Assignment Sheet (FILL OUT DURING MEETING)

29 POWER TEAM

- Owner: _____ Phone: _____ Email: _____
- Assistant 1: _____ Phone: _____ Email: _____
- Assistant 2: _____ Phone: _____ Email: _____

30 FLIGHT ELECTRONICS TEAM

- Owner: _____ Phone: _____ Email: _____
- Assistant 1: _____ Phone: _____ Email: _____
- Assistant 2: _____ Phone: _____ Email: _____

31 RESEARCH PAYLOAD TEAM

- Owner: _____ Phone: _____ Email: _____
- Assistant 1: _____ Phone: _____ Email: _____
- Assistant 2: _____ Phone: _____ Email: _____

32 INTEGRATION & TESTING TEAM

- Owner: _____ Phone: _____ Email: _____
- Assistant 1: _____ Phone: _____ Email: _____
- Assistant 2: _____ Phone: _____ Email: _____

ELECTRICAL LEAD (You)

- Name: _____ Phone: _____ Email: _____

Appendix B: Meeting #2 Preview (Week 1 Follow-Up)

32.1 Meeting #2 Agenda (1.5-2 hours)

- 1. Status Updates (40 min)
- Each subsystem: 5-min presentation + Q&A
- Demonstrate progress (show working subsystem)
- 2. Integration Planning (20 min)
- Review wiring diagram with actual components
- Plan wire routing and lengths
- Discuss mounting strategy for airframe
- 3. Hands-On Work (40 min)
- Continue subsystem testing
- Begin subsystem-to-subsystem interface testing
- Troubleshoot any issues discovered
- 4. Week 2 Planning (10 min)
- Define full system integration test plan
- Set date for full system bench test
- 5. Closing (10 min)
- Confirm Week 2 deliverables
- Address any concerns

Appendix C: Leadership Tips (For You)

32.2 Being a good technical leader

- BE CLEAR: Say exactly what needs to happen, by when, and who's responsible.
- BE CONFIDENT: You know more than your team. They're looking to you for guidance.
- BE SUPPORTIVE: When someone gets stuck, help them learn, don't just fix it.
- BE FIRM: If safety is compromised, shut it down immediately. No negotiations.
- BE ACCOUNTABLE: If the project fails, it's on you. Own it.
- DON'T micromanage: Give ownership, then get out of the way.
- DON'T be a bottleneck: Empower team to make decisions within their subsystems.
- DON'T ignore problems: Address issues immediately, don't let them fester.
- DON'T take all the work: Your job is to LEAD, not do everything yourself.

32.3 Remember

- This is a portfolio project for everyone, including you
- Your leadership will be more impressive than perfect technical work
- Employers want to see: systems thinking, communication, organization
- This project demonstrates all of that

You've got this.

33 DOCUMENT INFORMATION

Version: 1.0
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Status: Ready for Meeting

34 PRINT THIS DOCUMENT. BRING IT TO THE MEETING. FOLLOW IT.

Good luck. You're going to do great.